

# MUGDHA TADWALKAR

Bengaluru, Karnataka, India

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## Summary

Agentic AI Developer with **1.5+ years** designing and deploying **multi-agent systems, LLM-driven pipelines, and autonomous AI solutions** in enterprise environments. Hands-on with **Python, LangChain, CrewAI, RAG, Agentic Orchestration, and Vector Databases** – building agents capable of reasoning, planning, and executing tasks end-to-end. Proficient in **LLM Integration & Deployment, FastAPI, REST APIs, CI/CD, and cloud platforms (Azure/AWS)** with full-lifecycle ownership from POC to production deployment.

## Education

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| <b>JSPM's College of Engineering</b><br><i>B.E in Computer Science and Engineering</i> | <b>2020 – 2024</b><br><i>Pune, India</i> |
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## Experience

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| <b>Software Engineer – AI &amp; Data Solutions</b><br><i>Tata Consultancy Services (TCS)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>May 2025 – Present</b><br><i>Bengaluru, India</i>   |
| <ul style="list-style-type: none"><li>Developed AI-powered document extraction workflows for a <b>UK pension administration platform</b> using <b>Python</b>, automating the processing of pension documents including SOD, LOA, and related forms.</li><li>Engineered <b>multimodal LLM pipelines</b> using <b>Gemma 3/Gemini</b> with OCR integration for document classification and structured data extraction from scanned PDFs and email attachments.</li><li>Designed advanced <b>prompt engineering</b> strategies and JSON schema-driven extraction pipelines to accurately capture member, deceased, notifier, and death certificate information.</li><li>Implemented response validation, guardrails, and error-handling mechanisms to improve extraction consistency and reduce incorrect document classifications in production workflows.</li><li>Integrated AI services with enterprise APIs and backend components, contributing to scalable document-processing workflows for UK pension operations.</li><li>Collaborated with cross-functional teams in an <b>Agile/Scrum</b> environment to deliver production-ready AI features, perform UAT support, and continuously improve model performance through iterative experimentation.</li></ul> |                                                        |
| <b>Digital Solutions Intern</b><br><i>Expert Global Solutions</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Jan 2024 – Jul 2024</b><br><i>Aurangabad, India</i> |
| <ul style="list-style-type: none"><li>Developed <b>Python backend services</b> for analytics workflows using <b>SQL and MongoDB</b> – contributed to API integration, defect resolution, and Agile delivery.</li><li>Built and optimized <b>Azure cloud-based pipelines</b> for high-volume data processing – ensuring scalability, fault tolerance, and reliability under production load.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                        |

## Projects

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| <b>LeadCommand – Scalable Multi-Agent AI Platform</b><br><i>Python, FastAPI, CrewAI, LangChain, Groq/LLaMA, Supabase, Next.js, GitHub Actions</i>                                                                                                                                                                                                                                                                                                                               | <b>2025 – Present</b> |
| <ul style="list-style-type: none"><li>Architected a <b>production multi-agent AI platform</b> using <b>CrewAI</b> – designing <b>autonomous agents with tool-use, memory, and agentic orchestration</b> for real estate ad automation end-to-end.</li><li>Built <b>modular, independently deployable backend services</b> with FastAPI and CI/CD (GitHub Actions) – applying best practices in <b>guardrails, error handling, observability, and REST API design</b>.</li></ul> |                       |
| <b>RAG-Powered IT Helpdesk – Agentic NLP System</b><br><i>Python, FastAPI, ChromaDB, Groq/LLaMA, LangChain, TF-IDF, Streamlit</i>                                                                                                                                                                                                                                                                                                                                               | <b>2025</b>           |
| <ul style="list-style-type: none"><li>Built an <b>autonomous agent</b> combining <b>semantic RAG retrieval</b> (ChromaDB vector database) with an <b>NLP classifier</b> – implemented <b>fallback mechanisms and evaluation pipelines</b> to validate reliability before enterprise rollout.</li></ul>                                                                                                                                                                          |                       |

## Technical Skills

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| <b>Programming:</b> Python, SQL, Java, Pandas, NumPy                                                                                                                       |
| <b>Agentic AI &amp; Orchestration:</b> LangChain, CrewAI, LangGraph (fundamentals), AutoGen (fundamentals), Multi-Agent Systems, Agentic Orchestration, Prompt Engineering |
| <b>LLMs &amp; Integration:</b> LLM Integration & Deployment, RAG, LLaMA, GPT-4o, DeepSeek                                                                                  |
| <b>Vector Databases &amp; Memory:</b> ChromaDB, FAISS (fundamentals), Pinecone (fundamentals), Semantic Search, Embedding Pipelines                                        |
| <b>Backend &amp; Deployment:</b> FastAPI, REST APIs, Microservices, CI/CD (GitHub Actions), Docker (fundamentals), Azure, AWS                                              |
| <b>ML &amp; NLP:</b> Scikit-learn, XGBoost, NLP, Feature Engineering, Model Evaluation, Statistical Analysis                                                               |
| <b>Databases:</b> PostgreSQL, MySQL, MongoDB, Supabase, ChromaDB                                                                                                           |

## Certifications

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| Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |
| Problem Solving (Intermediate) – HackerRank                         |